

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

LAB PROJECT REPORT



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Project Title	:	Movie Ticket Booking System
Course Name	:	Object Oriented Programming through Java Lab
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CERTIFICATE

This is to certify that the Object Oriented Programming through Java Lab project report entitled “Movie Ticket Booking System” submitted by **Challawar Gurupal (25915A1204)** to Vidya Jyothi Institute of Technology (An Autonomous Institution), Hyderabad, of B. Tech. II year, II Semester in Information Technology a Bonafide record of Lab project work carried out by them under my supervision.

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Signature of Head of the Department

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Professor

Abstract

The Movie Ticket Booking System is a Java-based desktop application developed to automate the process of booking movie tickets and managing theatre seat reservations. Traditional ticket booking methods require manual seat allocation and payment processing, which can consume significant time and may lead to booking errors. This project provides a computerized solution that allows users to select movies, choose seats, book tickets, and simulate payments efficiently.

The main objective of the project is to create a secure, efficient, and user-friendly movie ticket booking platform using Java technologies. The system includes features such as user login authentication, movie selection, seat reservation, payment simulation, ticket generation, and booking history management. The application uses Java Swing for graphical user interface development, JDBC for database connectivity, and MySQL for data storage.

The project also demonstrates important Real-Time Java concepts such as Collections, Multithreading, Exception Handling, Object-Oriented Programming, GUI programming, and Database Connectivity. The multithreading module helps manage simultaneous bookings and timer-based seat holding functionality. Collections are used for storing movie details, seat information, and booking records dynamically.

The expected outcome of the project is an efficient ticket booking platform that improves theatre management, reduces paperwork, saves time, and provides quick and accurate booking services. The system can be further enhanced with online payment gateways, mobile application support, and cloud integration for large-scale usage.

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Chapter 1 – Introduction

1.1 Project Overview

The Movie Ticket Booking System is designed to provide digital ticket booking services through a computer-based application. The system allows users to log in, view available movies, select seats, make bookings, and receive booking confirmation instantly.

Administrators can add movies, manage show timings, and monitor booking details.

The project reduces manual work involved in managing ticket reservations and seat allocation. It provides a fast, secure, and reliable method for movie ticket booking. The seat management functionality ensures that duplicate bookings are avoided and seats are allocated correctly.

Real World Applications:

- Multiplex theatres
- Cinema booking platforms
- Entertainment ticket booking systems
- Online event reservation systems

1.2 Objectives

- To automate the movie ticket booking process
- To reduce manual booking errors
- To provide quick ticket confirmation
- To improve efficiency and accuracy
- To implement Java real-time concepts practically
- To maintain secure customer booking records

1.3 Scope of the Project

The system is mainly used by:

- Customers
- Theatre Administrators
- Booking Managers

Future Scalability

Web-based deployment

Mobile application support

Online payment gateway integration

Cloud database integration

Chapter 2 – System Analysis

2.1 Functional Requirements

The system should perform the following functions:

- User login authentication
- Display available movies and show timings
- Seat selection and reservation
- Ticket booking confirmation
- Payment simulation
- Booking history generation
- Database storage of customer records
- Admin movie management

2.2 Non-Functional Requirements

Security

- Password protection
- Authorized access only

Performance

- Fast response during booking
- Efficient database connectivity

Reliability

- Accurate seat allocation

- Stable system execution

Usability

- Simple graphical user interface
- Easy navigation

2.3 Software Requirements

Requirement Details

OS	Windows/Linux
Language	Java
IDE	NetBeans/Eclipse
Database	MySQL
JDK	JDK 8 or above

2.4 Hardware Requirements

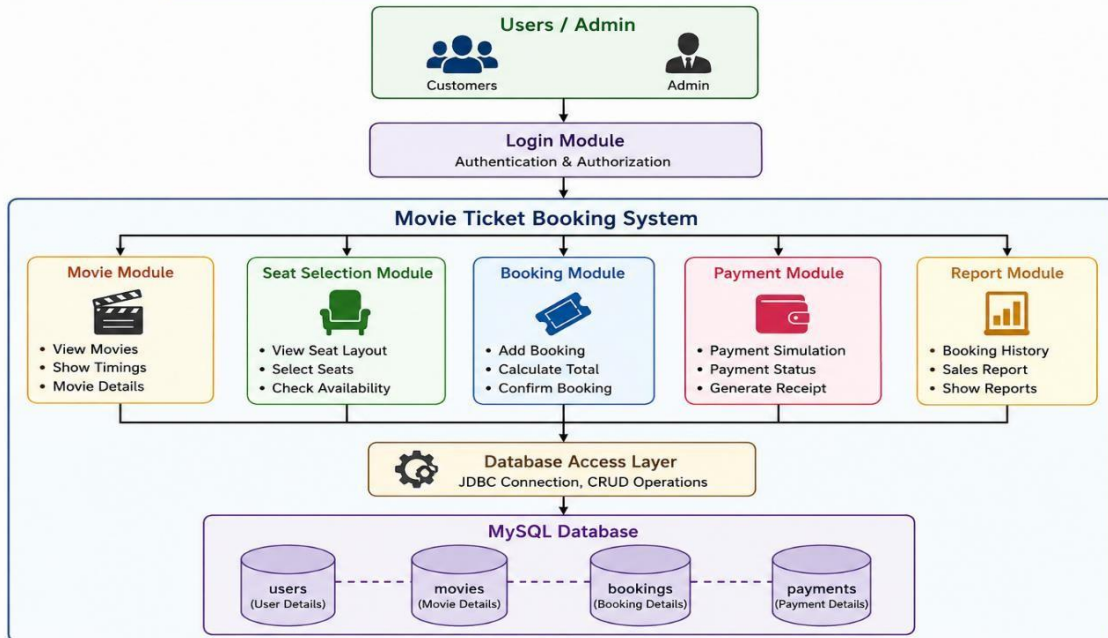
Requirement Details

RAM	4 GB
Processor	Intel i3 or above
Storage	500 GB

Chapter 3 – System Design

3.1 Architecture Diagram

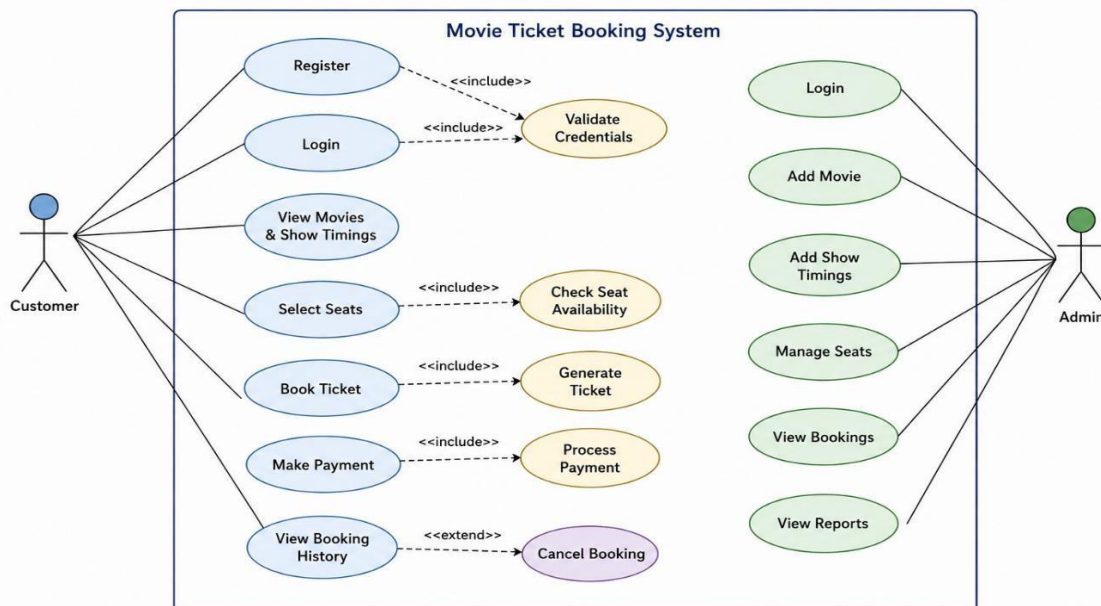
Movie Ticket Booking System – Architecture Diagram



3.2 UML Diagrams

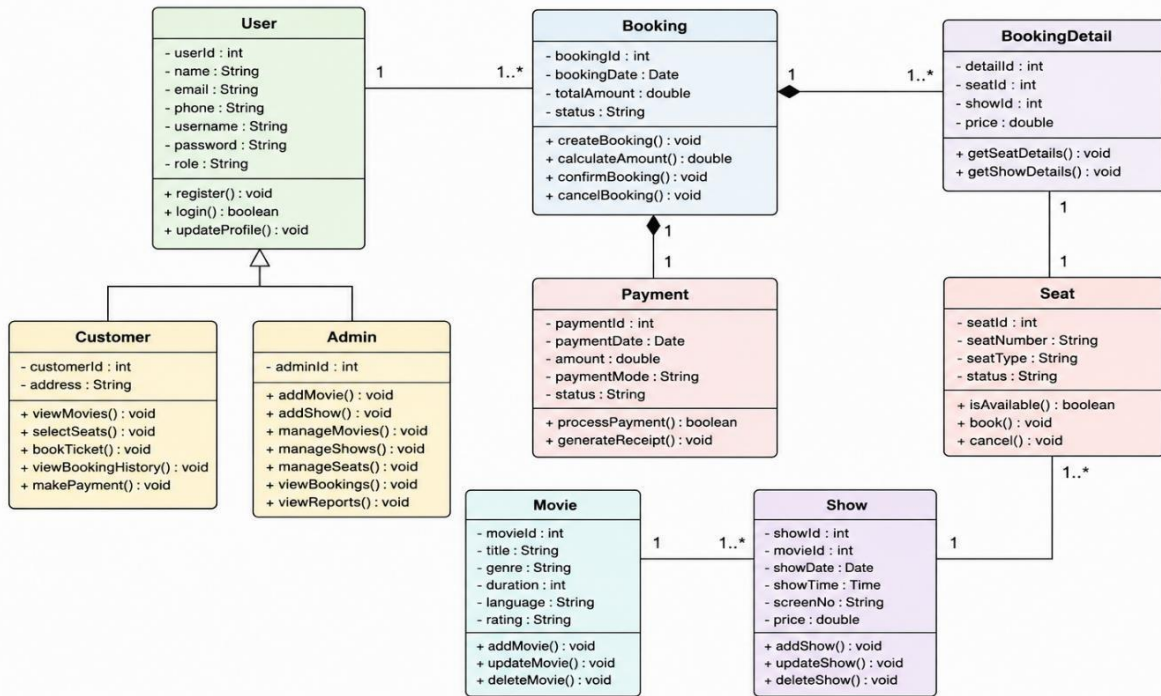
- Use Case Diagram

Use Case Diagram – Movie Ticket Booking System



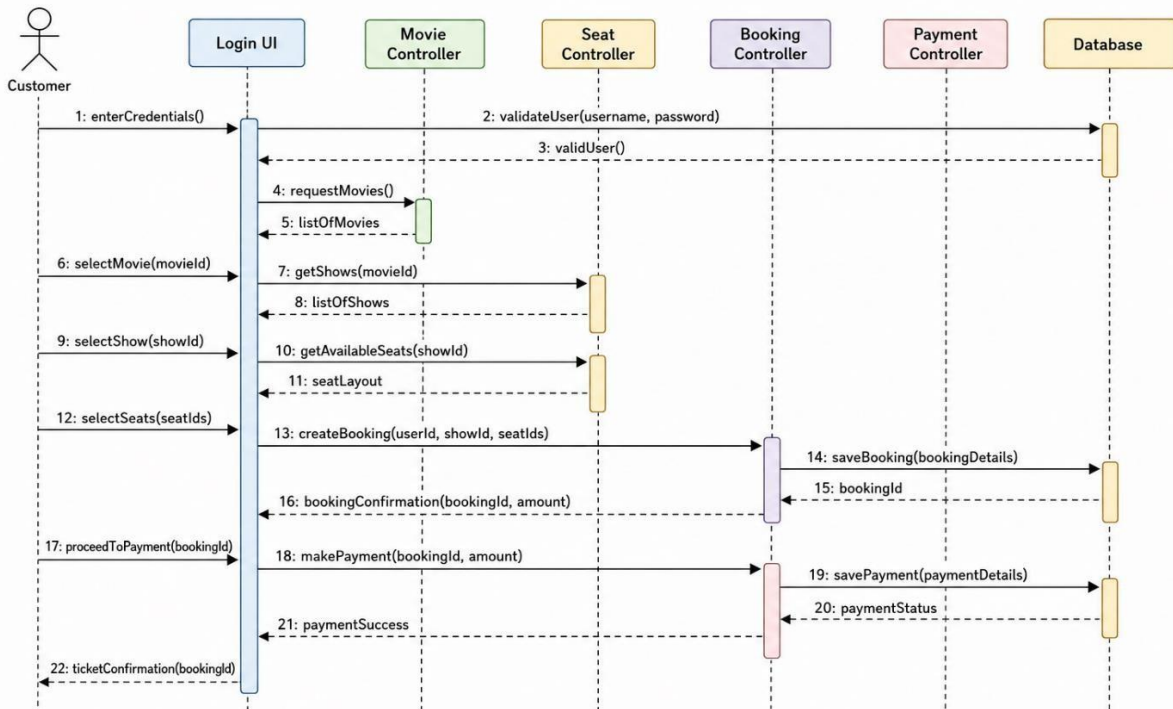
- Class Diagram

Class Diagram – Movie Ticket Booking System



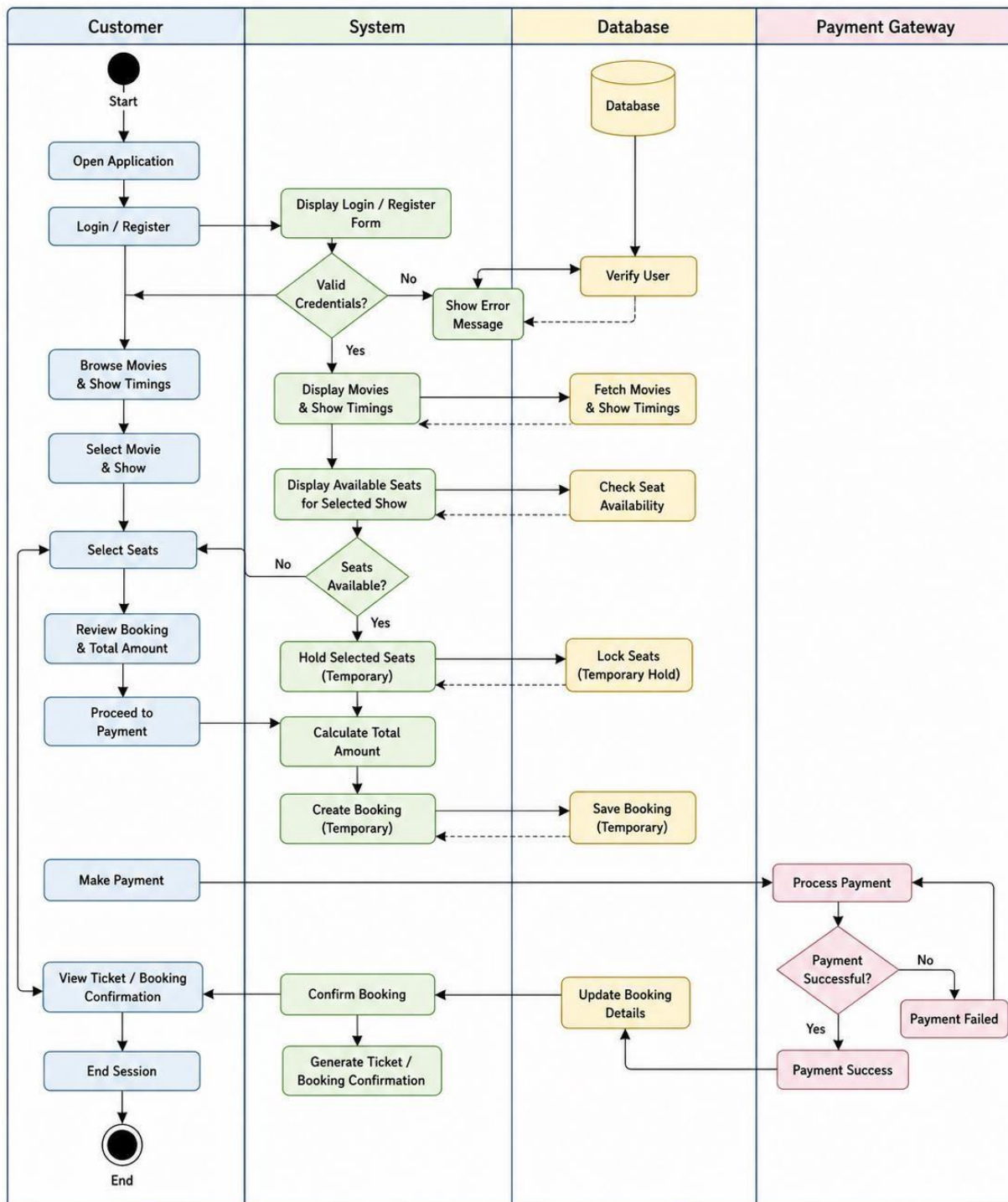
- Sequence Diagram

Sequence Diagram – Movie Ticket Booking System



- Activity Diagram

Activity Diagram – Movie Ticket Booking System



3.3 Database Design

Table Name	Purpose
users	Store customer login details
movies	Store movie details
bookings	Store ticket booking details
payments	Store payment records

Chapter 4 – Implementation

4.1 Modules Description

Module	Description
Login Module	User authentication
Movie Module	Display available movies
Seat Selection Module	Seat reservation management
Booking Module	Ticket booking process
Payment Module	Payment simulation
Report Module	Generate booking reports

4.2 Important Java Concepts Used

The project uses the following Java concepts:

- Classes and Objects
- Inheritance
- Polymorphism
- Exception Handling
- Multithreading
- Collections
- JDBC
- AWT/Swing GUI

4.3 Code Snippets

Database Connection Code:

```
import java.sql.Connection;

import java.sql.DriverManager;

public class DBConnection {

    Connection con;

    public Connection getConnection() {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/examdb",

                "root",

                "password");

        } catch (Exception e) {

            System.out.println(e);

        }

        return con;

    }

}
```

Seat Booking Code:

```
import java.util.ArrayList;

class SeatBooking {

    ArrayList<String> bookedSeats = new ArrayList<>();

    public void bookSeat(String seatNo) {

        if(!bookedSeats.contains(seatNo)) {

            bookedSeats.add(seatNo);

            System.out.println("Seat Booked: " + seatNo);

        } else {

            System.out.println("Seat Already Reserved");

        }

    }

}
```

Timer Thread Code:

```
class BookingTimer extends Thread {  
  
    int time = 30;  
  
    public void run() {  
  
        try {  
  
            while(time > 0) {  
  
                System.out.println("Time Left: " + time);  
  
                time--;  
  
                Thread.sleep(1000);  
  
            }  
  
        } catch(Exception e) {  
  
            System.out.println(e);  
  
        }  
  
    }  
  
}
```

Login Validation Code:

```
if(username.equals("admin") && password.equals("1234")) {  
  
    System.out.println("Login Successful");  
  
} else {  
  
    System.out.println("Invalid Credentials");  
  
}
```


Chapter 5 – Testing

5.1 Test Cases

Test Case ID	Input	Expected Output	Result
TC01	Valid Login	Login Success	Pass
TC02	Invalid Password	Error Message	Pass
TC03	Select Available Seat	Seat Reserved	Pass
TC04	Duplicate Seat Booking	Booking Rejected	Pass
TC05	Payment Success	Ticket Generated	Pass

5.2 Error Handling

The following exceptions are handled in the project:

- NumberFormatException
- SQLException
- NullPointerException
- ArrayIndexOutOfBoundsException

Chapter 6 – Results and Screenshots

Figure 1 – Login Screen

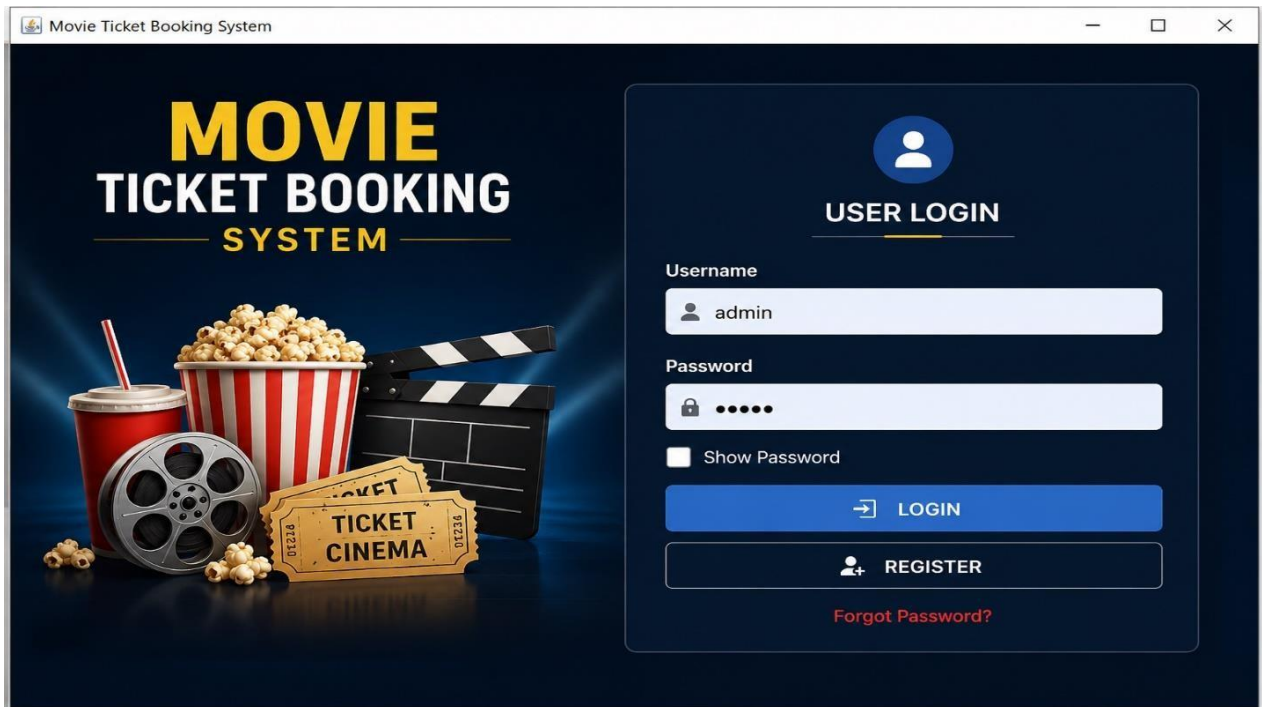


Figure 2 – Main Menu

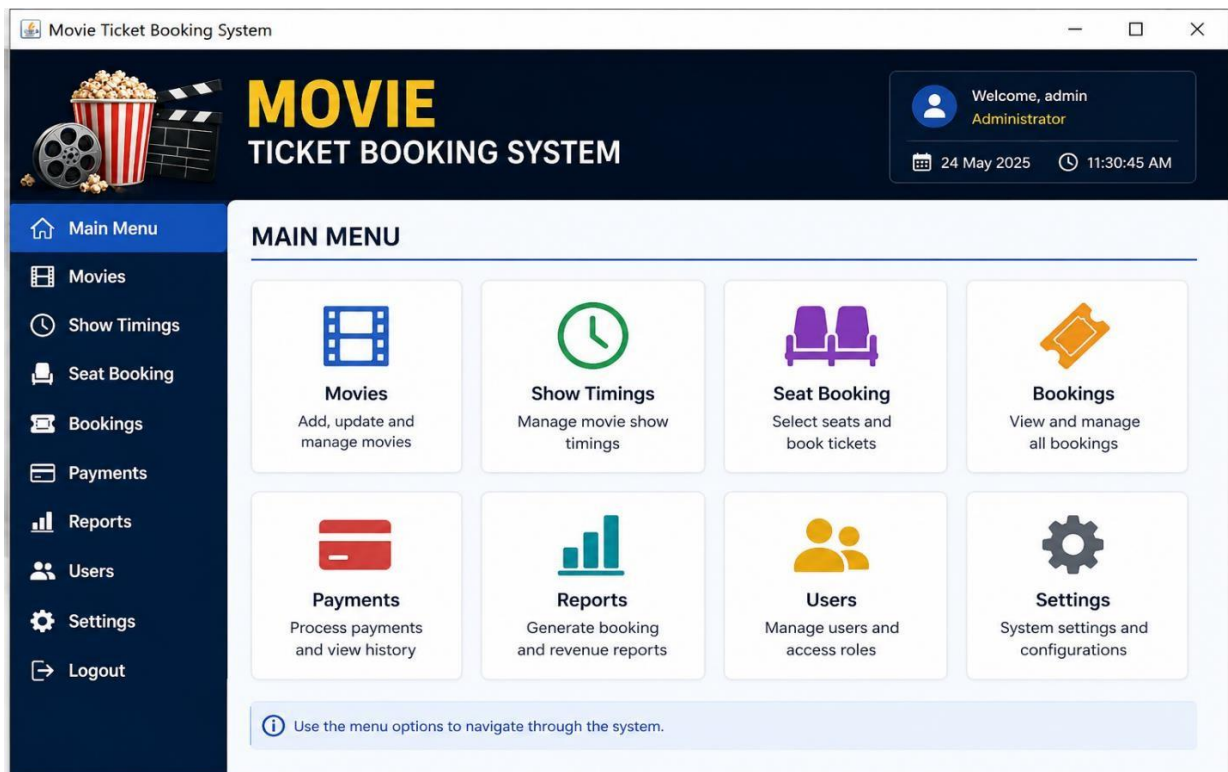


Figure 3 – Seat Selection Screen

The screenshot shows the 'SEAT SELECTION' interface for the 'Avengers: Endgame' movie on May 24, 2025, at 06:30 PM on Screen 1. The seat chart is a 10x10 grid (rows A-J, columns 1-10). Seats are color-coded: green for 'Available', grey for 'Booked', and white for 'Selected'. In row E, columns 6 and 7 are selected. The right sidebar contains a 'BOOKING SUMMARY' (Movie: Avengers: Endgame, Date: 24 May 2025, Time: 06:30 PM, Screen: Screen 1), 'SELECTED SEATS (2)' (E6, E7), and totals: Total Seats: 2, Price per Seat: ₹ 200.00, Total Amount: ₹ 400.00. Buttons for 'Confirm Booking' and 'Cancel Selection' are at the bottom right.

MOVIE TICKET BOOKING SYSTEM

Welcome, admin Logout

SEAT SELECTION

Movie: Avengers: Endgame Date: 24 May 2025 Time: 06:30 PM Screen: Screen 1

SCREEN

A	Available	Available	Available	Available	Available	Available	Available	Available	Available	A
B	Available	Available	Available	Available	Available	Available	Available	Available	Available	B
C	Available	Available	Booked	Booked	Booked	Booked	Booked	Booked	Available	C
D	Available	Available	Available	Available	Available	Available	Available	Available	Available	D
E	Available	Available	Available	Available	Selected	Selected	Available	Available	Available	E
F	Available	Available	Available	Available	Available	Available	Available	Available	Available	F
G	Available	Available	Booked	Booked	Booked	Booked	Booked	Booked	Available	G
H	Available	Available	Available	Available	Available	Available	Available	Available	Available	H
I	Available	Available	Available	Available	Available	Available	Available	Available	Available	I
J	Available	Available	Available	Available	Available	Available	Available	Available	Available	J

Available Booked Selected

BOOKING SUMMARY

Movie: Avengers: Endgame
Date: 24 May 2025
Time: 06:30 PM
Screen: Screen 1

SELECTED SEATS (2)

E6 x E7 x

Total Seats: 2
Price per Seat: ₹ 200.00
Total Amount: ₹ 400.00

Confirm Booking Cancel Selection

Figure 4 – Payment Screen

The screenshot shows the 'PAYMENT' interface. A green message states 'Seats selected successfully! Please proceed with payment.' The left sidebar shows the 'Payments' menu item selected. The main area has two sections: 'SELECT PAYMENT METHOD' with options for Credit / Debit Card, Net Banking, UPI, and Wallet; and 'PAYMENT DETAILS' with fields for Card Number (1234 5678 9012 3456), Expiry Date (MM / YY), CVV (123), and Card Holder Name (JOHN DOE). A note indicates this is a payment simulation. Buttons for 'Proceed to Pay ₹ 400.00' and 'Cancel' are present. The right sidebar contains a 'BOOKING SUMMARY' and 'Selected Seats' (E6, E7) with the same totals as Figure 3. A security badge at the bottom right states '100% Secure Payment'.

MOVIE TICKET BOOKING SYSTEM

Welcome, admin Logout

PAYMENT

✓ Seats selected successfully! Please proceed with payment.

SELECT PAYMENT METHOD

☒ Credit / Debit Card
Pay using your card

☐ Net Banking
Pay using net banking

☐ UPI
Pay using any UPI app

☐ Wallet
Pay using digital wallet

PAYMENT DETAILS

Card Number: 1234 5678 9012 3456

Expiry Date: MM / YY CVV: 123

Card Holder Name: JOHN DOE

This is a payment simulation. No real payment will be processed.

Proceed to Pay ₹ 400.00 Cancel

BOOKING SUMMARY

Movie : Avengers: Endgame
Date : 24 May 2025
Time : 06:30 PM
Screen : Screen 1

Selected Seats

E6 E7

Price per Seat : ₹ 200.00
Total Seats : 2
Total Amount : ₹ 400.00

100% Secure Payment
Your payment information is safe and encrypted.

Figure 5 – Booking Confirmation Screen

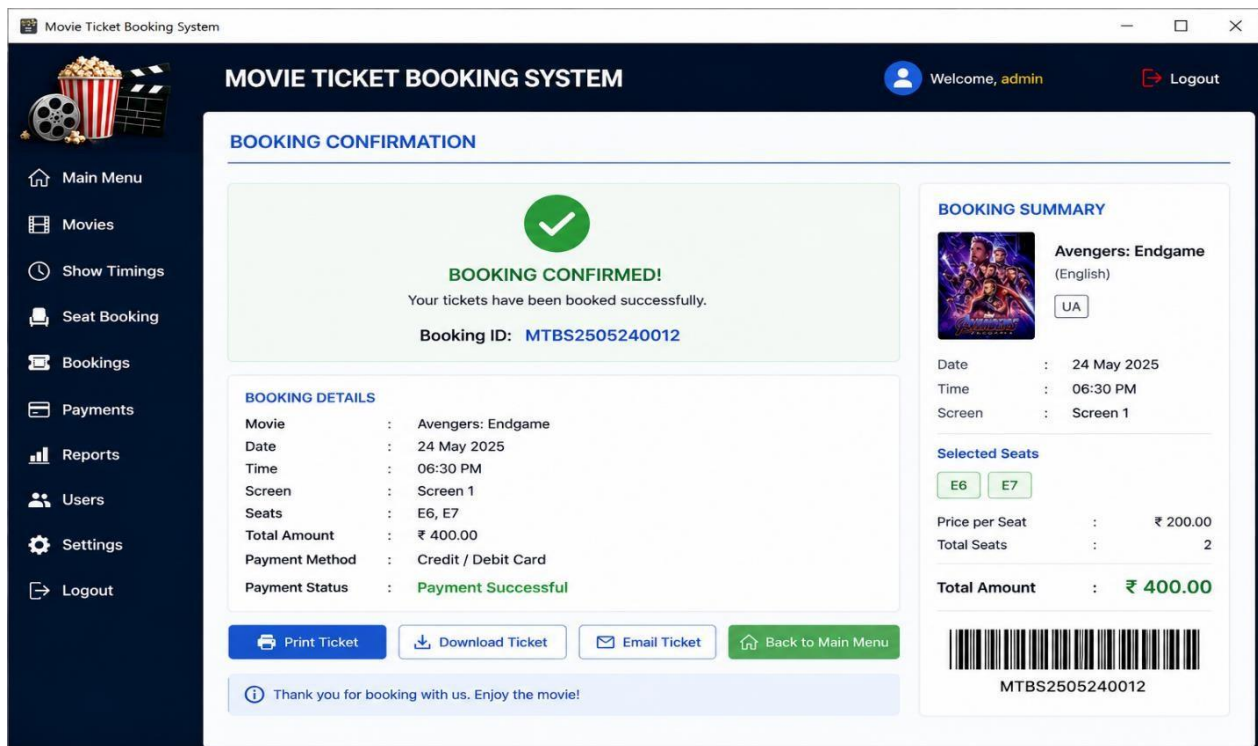
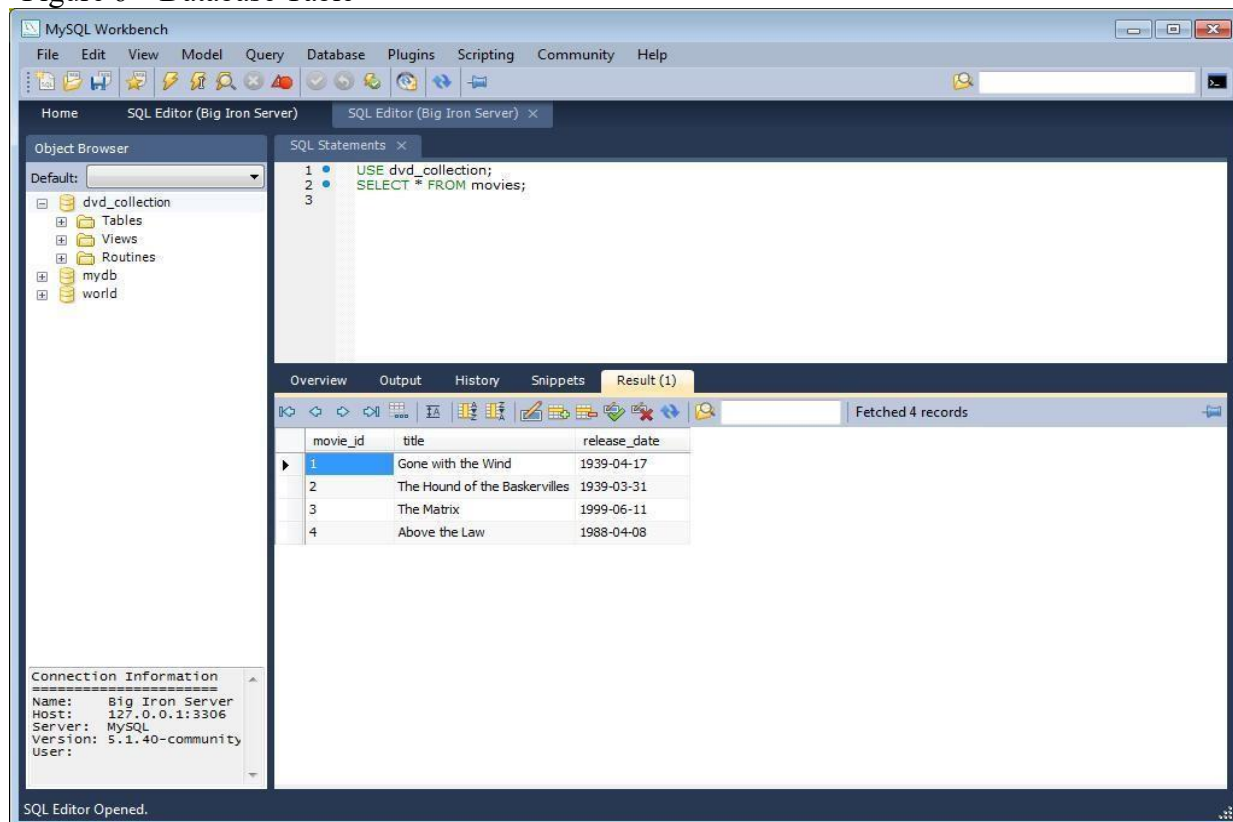


Figure 6 – Database Table



Chapter 7 – Conclusion and Future Scope

Conclusion

The Movie Ticket Booking System was successfully developed using Java and MySQL technologies. The project automates the ticket reservation process and reduces manual effort involved in managing bookings and seat allocation. The system provides secure login authentication, efficient seat management, payment simulation, and instant ticket confirmation. This project helped in understanding important Real-Time Java concepts such as multithreading, collections framework, exception handling, GUI development, and JDBC connectivity. The system improves efficiency, accuracy, and reliability in theatre booking management.

Future Enhancements

- Online payment gateway integration
- Mobile application support
- Cloud database integration
- AI-based movie recommendation system
- Web-based deployment
- QR code-based ticket verification

References

1. Java: The Complete Reference
2. Head First Java
3. MySQL Workbench
4. Oracle Java Documentation
5. Java Swing Documentation